Houston City College

ANALYSIS REPORT

Annual Medical Cost Prediction Analysis

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Course: ITAI-1371 - Machine Learning
Assignment: Final Project – Analysis Report
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1. Introduction & Approach

The goal of this project was to build a machine learning model that can predict how much patients will spend on medical costs annually. I decided to tackle this as a regression problem, using a clean dataset that we originally prepared during the Midterm. My approach was to start with simple models and then move to more complex ones, like ensembles, to see if I could get better accuracy.

2. Methodology & Feedback Implementation.

One of the most important parts of this final stage was fixing the issues we had in the Midterm. I followed the professor's feedback closely:

- Handling Outliers: I noticed that some patients had extremely high costs (over $40k or $50k). These were skewing the results, so I filtered the data to only include costs up to $30,000. This made the models much more stable.
- Log Transformation: Since medical costs are usually "skewed" (most people pay a little, a few pay a lot), I used the log1p transformation on the target variable. This normalized the data and helped the models learn better.
- Data Cleaning: I also removed the bmi_smoker_interaction feature because we found an error in how it was calculated before.

3. Model Discussion.

I tested five different models: Linear Regression, Decision Tree, Random Forest, XGBoost, and KNN.

- The tree-based models (Random Forest and XGBoost) performed the best because they are

good at picking up non-linear relationships, like how being a smoker affects people differently depending on their BMI.

- Linear Regression and Bayesian models were okay but a bit too simple for this data.
- The Voting Regressor ended up being the "Champion" because it averages the best models, which helps reduce the chance of the model being wrong on a specific case.

4. Final Results

My final model achieved an **R^2 score of 0.81** on the Test set. This is a very good result because it means the model can explain about 81% of the variation in costs. Also, the MAE was around **$2,518**, which is a reasonable margin of error for medical expenses.

5. Conclusion

The project shows that factors like smoking, BMI, and age are huge predictors for medical costs. By cleaning the data correctly and using an ensemble model, I was able to create a tool that predicts these costs with high confidence.